

Oscillatory Motion

Oscillation

Oscillation refers to a **repetitive variation, fluctuation, or movement around an equilibrium point or between two points**. It is a fundamental concept in various scientific disciplines, including physics, mathematics, and engineering. **In the context of physics, oscillation typically involves the periodic back-and-forth motion of a physical system around its equilibrium position.**

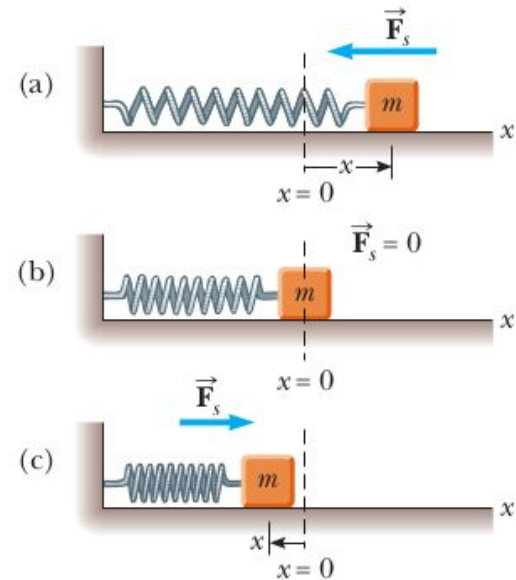
Oscillations can occur in various forms, such as mechanical oscillations (e.g., a swinging pendulum), electrical oscillations (e.g., alternating current in circuits), and even in the form of waves (e.g., sound waves, light waves, or water waves). The behavior of oscillatory systems can often be described mathematically using concepts from trigonometry and calculus, such as sine and cosine functions.

Motion of an object attached to a spring

Consider a block of mass m attached to the end of a spring, with the block free to move on a horizontal, frictionless surface

Hooke's law

$$F_s = -kx$$



Restoring force: F_s is called restoring force because it is always directed toward the equilibrium position and therefore opposite the displacement of the block from equilibrium.

$$-kx = ma_x$$

$$a_x = -\frac{k}{m}x$$

Simple Harmonic Motion

A Simple Harmonic Motion, or **SHM**, is defined as a motion in which the restoring force is directly proportional to the displacement of the body from its mean position. The direction of this restoring force is always towards the mean position.

An object moves with simple harmonic motion whenever its acceleration is proportional to its position and is oppositely directed to the displacement from equilibrium.

The Particle in Simple Harmonic Motion

$$a_x = -\frac{k}{m} x$$

That is, the acceleration of the block is proportional to its position, and the direction of the acceleration is opposite the direction of the displacement of the block from equilibrium. Systems that behave in this way are said to exhibit simple harmonic motion. An object moves with simple harmonic motion whenever its acceleration is proportional to its position and is oppositely directed to the displacement from equilibrium.

$$\frac{d^2x}{dt^2} = -\omega^2x$$

We need to find a function $x(t)$ that satisfies this second-order differential equation and is a mathematical representation of the position of the particle as a function of time. We seek a function whose second derivative is the same as the original function with a negative sign and multiplied by ω^2 . The trigonometric functions sine and cosine exhibit this behavior, so we can build a solution around one or both of them. The following cosine function is a solution to the differential equation:

$$x(t) = A \cos (\omega t + \phi)$$

A = Amplitude

Omega = angular frequency

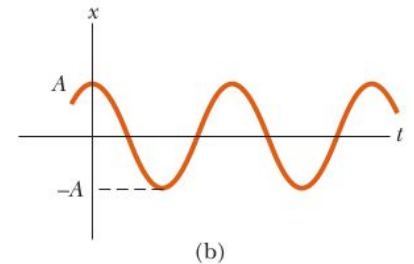
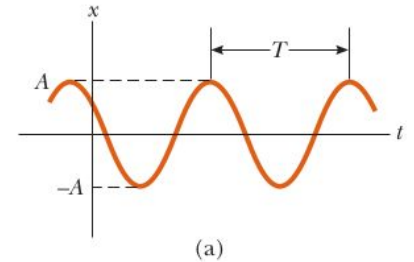
Theta = phase constant (or initial phase angle)

Period

The period T of the motion is the time interval required for the particle to go through one full cycle of its motion

$$[\omega(t + T) + \phi] - (\omega t + \phi) = 2\pi$$

$$T = \frac{2\pi}{\omega}$$



Frequency

The inverse of the period is called the frequency f of the motion. Whereas the period is the time interval per oscillation, the frequency represents the number of oscillations the particle undergoes per unit time interval:

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

the period and frequency depend only on the mass of the particle and the force constant of the spring and not on the parameters of the motion

Velocity and acceleration of a particle undergoing simple harmonic motion

$$v = \frac{dx}{dt} = -\omega A \sin (\omega t + \phi)$$

$$a = \frac{d^2x}{dt^2} = -\omega^2 A \cos (\omega t + \phi)$$

Maximum velocity and acceleration

$$v_{\max} = \omega A = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = \omega^2 A = \frac{k}{m} A$$

Damped Oscillation

Damped oscillation refers to the oscillatory motion of a system that gradually loses energy over time due to the presence of damping forces, such as friction, air resistance, or other dissipative forces. As a result, the amplitude of the oscillations decreases over time, eventually leading to the system coming to rest at its equilibrium position.

Forced oscillation

Forced oscillation refers to the oscillatory motion of a system that is subject to an external periodic driving force or input. The system oscillates at the frequency of the external force, which can be different from the natural frequency of the system. When the frequency of the external force matches the natural frequency of the system, resonance can occur, leading to amplified oscillations.

In the context of forced oscillation, the amplitude of the oscillations depends on the frequency of the driving force, the natural frequency of the system, and the damping present in the system. If the frequency of the external force is close to the natural frequency of the system, the amplitude of the oscillations can be significantly higher compared to when the frequencies are mismatched.